New File at / - charulatha122/char | 1-Finding Duplicates-O(n^2) Time | New tab
File C:/Users/Charulatha/Downloads/1-Finding%20Duplicates-O(n%5E2)%20Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

1-Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity

Started on	Tuesday, 29 September 2026, 3:34 PM
State	Finished
Completed on	Tuesday, 29 September 2026, 3:42 PM
Time taken	8 mins 25 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

25°C Mostly cloudy
Search
ENG IN 78% 10:07 PM 02-11-2026

New File at / - charulatha122/char | 1-Finding Duplicates-O(n^2) Time | New tab
File C:/Users/Charulatha/Downloads/1-Finding%20Duplicates-O(n%5E2)%20Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

Question 1 | Correct | Mark 1.00 out of 1.00 | Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(int i=0;i<n;i++)
    {
        for(int j=i+1;j<n;j++)
        {
            if(a[i]==a[j])
```

25°C Mostly cloudy
Search
ENG IN 78% 10:07 PM 02-11-2026

New File at / - charulatha122/chari | 1-Finding Duplicates-O(n^2) Time | New tab
File C:/Users/Charulatha/Downloads/1-Finding%20Duplicates-O(n%5E2)%20Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

```c
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(int i=0;i<n;i++)
    {
        for(int j=i+1;j<n;j++)
        {
            if(a[i]==a[j])
            {
                printf("%d",a[i]);
                break;
            }
        }
    }
}
```

	Input	Expected	Got	
☐	11 10 9 7 6 5 1 2 3 8 4 7	7	7	☐
☐	5 1 2 3 4 4	4	4	☐
☐	5 1 1 2 3 4	1	1	☐

Passed all tests! ☐

Correct

25°C Mostly cloudy
Search
ENG IN 78% 10:08 PM 02-11-2026

2-Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

Started on	Tuesday, 29 September 2026, 3:28 PM
State	Finished
Completed on	Tuesday, 29 September 2026, 3:33 PM
Time taken	5 mins 21 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

New File at / - charulatha122/chan | 1-Finding Duplicates-O(n^2) Time | New tab | 2-Finding Duplicates-O(n) Time C
File C:/Users/Charulatha/Downloads/2-Finding%20Duplicates-O(n)%20Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

Output **Format:**

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main()
{
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    int b[n];
    for(int i=0;i<n;i++)
    {
        b[i]=0;
    }
    for(int i=0;i<n;i++)
    {
        if(b[a[i]]==1)
        {
            printf("%d",a[i]);
            break;
        }
        b[a[i]]=1;
    }

```

Show hidden icons
25°C Mostly cloudy
Search
ENG IN
78%
10:08 PM 02-11-2026

New File at / - charulatha122/char | 1-Finding Duplicates-O(n^2) Time | New tab | 2-Finding Duplicates-O(n) Time C
File C:/Users/Charulatha/Downloads/2-Finding%20Duplicates-O(n)%20Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

```
    {
        b[i]=0;
    }
    for(int i=0;i<n;i++)
    {
        if(b[a[i]]==1)
        {
            printf("%d",a[i]);
            break;
        }
        b[a[i]]=1;
    }

}
```

	Input	Expected	Got	
☐	11 10 9 7 6 5 1 2 3 8 4 7	7	7	☐
☐	5 1 2 3 4 4	4	4	☐
☐	5 1 1 2 3 4	1	1	☐

Passed all tests! ☐

Correct

Marks for this submission: 1.00/1.00.

Finish review

25°C Mostly cloudy
Search
ENG IN
78%
10:08 PM 02-11-2026

3-Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Complexity

Started on	Tuesday, 29 September 2026, 3:43 PM
State	Finished
Completed on	Tuesday, 29 September 2026, 4:01 PM
Time taken	18 mins 16 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

New File at / - charulatha122/char | 1-Finding Duplicates-O(n^2) Time | New tab | 2-Finding Duplicates-O(n) Time C | 3-Print Intersection of 2 sorted arr
File C:/Users/Charulatha/Downloads/3-Print%20Intersection%20of%202%20sorted%20arrays-O(m_n)Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

25°C Mostly cloudy
Search
ENG IN 77% 10:10 PM 02-11-2026

New File at / - charulatha122/char | 1-Finding Duplicates-O(n^2) Time | New tab | 2-Finding Duplicates-O(n) Time C | 3-Print Intersection of 2 sorted arr
File C:/Users/Charulatha/Downloads/3-Print%20Intersection%20of%202%20sorted%20arrays-O(m_n)Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main()
{
    int t;
    scanf("%d",&t);

    while(t!=0)
    {

        int n1,n2;
    scanf("%d",&n1);
    int a[n1];
    for(int i=0;i<n1;i++)
    {
        scanf("%d",&a[i]);
    }
    scanf("%d",&n2);
    int b[n2];
    for(int i=0;i<n2;i++)
    {
        scanf("%d",&b[i]);
    }

        for(int i=0;i<n1;i++)
        {
          for(int j=0;j<n2;j++)
          {
             if(a[i]==b[j])
             {
                 printf("%d ",a[i]);
             }
          }
        }
        t--;
    }
}
```

25°C Mostly cloudy
Search
ENG IN
77%
10:10 PM 02-11-2026

New File at / - charulatha122/char | 1-Finding Duplicates-O(n^2) Time | New tab | 2-Finding Duplicates-O(n) Time C | 3-Print Intersection of 2 sorted arr
File C:/Users/Charulatha/Downloads/3-Print%20Intersection%20of%202%20sorted%20arrays-O(m_n)Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

```
                {
                    printf("%d ",a[i]);
                }
            }
        }
        t--;
    }
}
```

	Input	Expected	Got	
	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	
	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

Finish review

25°C Mostly cloudy
Search
ENG IN
77%
10:10 PM 02-11-2026

4-Print Intersection of 2 sorted arrays-O(m+n)Time Complexity,O(1) Space Complexity

Started on	Tuesday, 29 September 2026, 4:03 PM
State	Finished
Completed on	Thursday, 8 October 2026, 8:27 AM
Time taken	8 days 16 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

New File at / - charulatha122/char | 1-Finding Duplicates-O(n^2) Time | 2-Finding Duplicates-O(n) Time C | 3-Print Intersection of 2 sorted arr | 4-Print Intersection of 2 sorted arr
File C:/Users/Charulatha/Downloads/4-Print%20Intersection%20of%202%20sorted%20arrays-O(m+n)Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int n1, n2;
        scanf("%d", &n1);

        int a[n1];
        for (int i = 0; i < n1; i++)
            scanf("%d", &a[i]);

        scanf("%d", &n2);

        int b[n2];
        for (int i = 0; i < n2; i++)
            scanf("%d", &b[i]);

        int i = 0, j = 0;

        while (i < n1 && j < n2) {
            if (a[i] == b[j]) {
                printf("%d ", a[i]);
                i++;
                j++;
            }
            else if (a[i] < b[j]) {
                i++;
            }
            else {
                j++;
            }
        }

        printf("\n");
    }
```

25°C Mostly cloudy
Search
ENG IN
77%
10:11 PM 02-11-2026

New File at / - charulatha122/char | 1-Finding Duplicates-O(n^2) Time | 2-Finding Duplicates-O(n) Time C | 3-Print Intersection of 2 sorted arr | 4-Print Intersection of 2 sorted arr
File C:/Users/Charulatha/Downloads/4-Print%20Intersection%20of%202%20sorted%20arrays-O(m+n)Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

```
40      return 0;
41  }
```

	Input	Expected	Got	
☐	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	☐
☐	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	☐

Passed all tests! ☐

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

25°C Mostly cloudy
Search
ENG IN
77%
10:11 PM 02-11-2026

5-Pair with Difference-O(n^2)Time Complexity,O(1) Space Complexity

Started on	Tuesday, 29 September 2026, 4:23 PM
State	Finished
Completed on	Tuesday, 29 September 2026, 4:31 PM
Time taken	8 mins 35 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

New File at / - charulatha122/ | 1-Finding Duplicates-O(n^2) | 2-Finding Duplicates-O(n) Tir | 3-Print Intersection of 2 sorte | 4-Print Intersection of 2 sorte | 5-Pair with Difference-O(n^2
File C:/Users/Charulatha/Downloads/5-Pair%20with%20Difference-O(n%5E2)Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main()
{
    int n,k;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    scanf("%d",&k);
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
```

25°C Mostly cloudy
Search
ENG IN
76%
10:12 PM 02-11-2026

New File at / - charulatha122/ | 1-Finding Duplicates-O(n^2) | 2-Finding Duplicates-O(n) Tir | 3-Print Intersection of 2 sorte | 4-Print Intersection of 2 sorte | 5-Pair with Difference-O(n^2
File C:/Users/Charulatha/Downloads/5-Pair%20with%20Difference-O(n%5E2)Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

```c
    scanf("%d",&k);
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
        {
            if(a[j]-a[i]==k && i!=j)
            {
                printf("%d",1);
                return 0;
            }
        }
    }
    printf("%d",0);
}
```

	Input	Expected	Got	
	3 1 3 5 4	1	1	
	10 1 4 6 8 12 14 15 20 21 25 1	1	1	
	10 1 2 3 5 11 14 16 24 28 29 0	0	0	
	10 0 2 3 7 13 14 15 20 24 25 10	1	1	

25°C Mostly cloudy
Search
ENG IN 76% 10:12 PM 02-11-2026

6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

Started on	Tuesday, 29 September 2026, 4:32 PM
State	Finished
Completed on	Thursday, 8 October 2026, 8:28 AM
Time taken	8 days 15 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

New File at / - charulath | 1-Finding Duplicates-O | 2-Finding Duplicates-O | 3-Print Intersection of 2 | 4-Print Intersection of 2 | 5-Pair with Difference (| 6-Pair with Difference
File C:/Users/Charulatha/Downloads/6-Pair%20with%20Difference%20-O(n)%20Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n, k;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &k);

    int i = 0, j = 1;

    while (j < n)
    {
        int diff = a[j] - a[i];
```

25°C Mostly cloudy
Search
ENG IN 76% 10:13 PM 02-11-2026

```c
#include <stdio.h>

int main()
{
    int n, k;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &k);

    int i = 0, j = 1;

    while (j < n)
    {
        int diff = a[j] - a[i];

        if (i != j && diff == k)
        {
            printf("1");
            return 0;
        }
        else if (diff < k)
        {
            j++;
        }
        else
        {
            i++;

            if (i == j)
                j++;
        }
    }

    printf("0");

```

New File at / - charulath | 1-Finding Duplicates-O | 2-Finding Duplicates-O | 3-Print Intersection of 2 | 4-Print Intersection of 2 | 5-Pair with Difference-(| 6-Pair with Difference
File C:/Users/Charulatha/Downloads/6-Pair%20with%20Difference%20-O(n)%20Time%20Complexity,O(1)%20Space%20Complexity_%20Attempt%20review%20(1).mht
Chat

	Input	Expected	Got	
	3 1 3 5 4	1	1	
	10 1 4 6 8 12 14 15 20 21 25 1	1	1	
	10 1 2 3 5 11 14 16 24 28 29 0	0	0	
	10 0 2 3 7 13 14 15 20 24 25 10	1	1	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00

Finish review

Back to Course

25°C Mostly cloudy
Search
ENG IN
76%
10:13 PM 02-11-2026